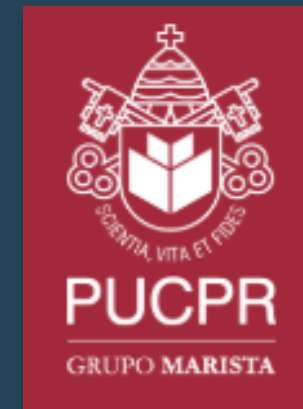


AULA 03

MÉTODOS DE PESQUISA EM COMPUTAÇÃO



Agenda

- Atividade 6 – Artigo RSL
 - Quiz Menti.com
- Visão geral
- Fases
- Condução
 - Quiz Menti.com
- Ferramentas
- Atividade 7 – Início mini pesquisa
- Próxima aula

ATIVIDADE 6:

Seguir os passos para busca de artigos sobre Revisão Sistemática de Leitura em pelo menos **duas** bases científica. Selecionar um dos artigos e salvar os resultados.

CONDUÇÃO:

Individual

DURAÇÃO:

10 minutos

[Full Text: revisão sistemática de literatura] AND [All: kitchenham] AND [All: computação] AND [Publication Date: (01/01/2008 TO 12/31/2018)]

Search anything within the ACM Digital Library or go to your [Saved Searches](#)

Search items from:

The ACM Full-Text collection



Search Within

Full text



revisão sistemática de literatura

Anywhere



Kitchenham

Anywhere



computação

QUIZ
MENTI.COM



Revisão Sistemática da Literatura (RSL)

- De acordo com (Kitchenham, 2004):
 - Tem como objetivo identificar, avaliar e interpretar toda a pesquisa relevante disponível para responder a uma questão de pesquisa, ou tópico de uma área, ou fenômeno de interesse.
 - Estudos individuais que contribuem para realizar a RSL são chamados estudos primários.
 - A RSL é uma forma de estudo secundário.

Revisão Sistemática da Literatura (RSL)

“Uma Revisão Sistemática da Literatura é uma forma de estudo secundário que utiliza um método bem definido para identificar, analisar e interpretar todas as evidências disponíveis relacionadas a um questão de pesquisa específica de uma forma não enviesada e (até certo ponto) repetível.”

Kitchenham, B. Guidelines for Performing Systematic Reviews in Software Engineering. EBSE Technical Report. 2007

Classificação dos estudos



Estudo primário: pesquisa original



Estudo secundário: estudos sobre estudos primários



Estudo terciário: estudo sobre os estudos secundários

Exemplo de estudo terciário

- Context: The concept of technical debt (TD) contextualizes problems faced during software evolution considering the tasks that are not carried out adequately during its development. Currently, it is common to associate any impediment related to the software product and its development process to the definition of TD. This can bring confusion and ambiguity in the use of the term. Besides, due to the increasing amount of work in the area, it is difficult to have a comprehensive view of the plethora of proposals on TD management.
- Objective: This paper intends to investigate the current state of research on TD by identifying what research topics have been considered, organizing research directions and practical knowledge that has already been defined, identifying the known types of TD, and organizing what activities, strategies and tools have been proposed to support the management of TD.
- **Method: A tertiary study was performed based on a set of five research questions. In total, 13 secondary studies, dated from 2012 to March 2018, were evaluated.**
- Results: The results of this tertiary study are beneficial for both practitioners and researchers. We evolved a taxonomy of TD types, identified a list of situations in which debt items can be found in software projects, and organized a map representing the state of the art of activities, strategies and tools to support TD management. Besides, we also summarized some research directions and practical knowledge, and identified the research topics that have been more considered in secondary studies.
- Conclusion: This tertiary study revisited the TD landscape. Its results can help to identify points that still require further investigation in TD research.

Rios, N. Mendonça, M.; Spindola, R. A tertiary study on technical debt: Types, management strategies, research trends, and base information for practitioners. *Information and Software Tehcnology*, 2018, v. 102, p. 117-145

Revisão Sistemática da Literatura (RSL)

- Há vários protocolos para RSL
- Tabela 10, pág. 27 (Kitchenham, 2004) apresenta algumas diferenças entre os vários processos de RSL
- Kitchenham (2004) tem como base:
 - The Cochrane Reviewer's Handbook
 - Guidelines – Australian National Health, Medical Research Council
 - CRD Guidelines

Revisão Sistemática da Literatura (RSL)

- Por que realizar uma RSL ?
 - Resumir evidências que existem sobre um tratamento ou tecnologia.
 - Identificar lacunas nas pesquisas realizadas para sugerir áreas para investigações futuras.
 - Fornecer uma revisão para posicionar/justificar novas pesquisas.

Revisão Sistemática da Literatura (RSL)

- Algumas diferenças entre RSL e Revisão “tradicional”
 - Requer **mais esforço** para ser realizada do que as revisões tradicionais.
 - RSL precisa **seguir um protocolo** para a revisão, com **questões de pesquisa e métodos** que serão utilizados para realizar a pesquisa.
 - RSL são baseadas em **estratégias de pesquisa definidas** que objetivam detectar a maior quantidade de literatura relevante.

Revisão Sistemática da Literatura (RSL)

- Algumas diferenças entre RSL e Revisão “tradicional” (cont.)
 - RSL **documentam** a estratégia de pesquisa de maneira que é possível replicar o estudo.
 - RSL requer **critérios de inclusão e exclusão** para selecionar os estudos potenciais.
 - RSL **especifica a informação a ser obtida** de cada estudo primário incluindo critérios de qualidade para avaliar cada um dos estudos.

Revisão Sistemática da Literatura (RSL)

- Onde utilizar ?
 - Incluir em um relatório técnico
 - Incluir como uma seção ou capítulo em uma dissertação de mestrado ou tese de doutorado
 - Publicar com estudo completo em journal ou conferência

ORGANIZATIONAL LEARNING APPLIED TO SOFTWARE ENGINEERING: A SYSTEMATIC REVIEW

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Received (Day Month Year)

Revised (Day Month Year)

Accepted (Day Month Year)

EXEMPLO DE RSL

Exemplo de RSL

Organizational learning assists the companies to significantly improve their processes by means of experiences reuse, making knowledge accessible to the whole organization. In the software engineering area it is important that the acquired knowledge is stored and systematically reused. This paper aims to present a systematic review, by identifying in which software engineering areas are the organizational learning studies concentrated, and how the organizational learning concepts are being applied in the software engineering area. This systematic review identified 2496 papers. After eliminating the duplicate titles and the ones not related to the review, remained 1184 papers. Applying the exclusion criteria, the number of papers was reduced to 68. These papers were analyzed and classified according to the software engineering areas defined in the SWEBOK, and the main organizational learning theories and techniques. It was observed that many software engineering researches apply organizational learning concepts without being aware of it.

Keywords: software engineering; organizational learning; systematic review; learning organization

Exemplo de RSL

applied in the software engineering area. This systematic review identified 2496 papers. After eliminating the duplicate titles and the ones not related to the review, remained 1184 papers.

Menolli, A.; Reinehr, S.; Malucelli, A. Organizational Learning Applied to Software Engineering: a Systematic Review. International Journal of Software Engineering and Knowledge Engineering. v.23

Exemplo de RSL



Contents lists available at [SciVerse ScienceDirect](#)

Expert Systems with Applications

journal homepage: www.elsevier.com/locate/eswa



Review

Ontologies application in organizational learning: A literature review

Joselaine Valaski*, Andreia Malucelli, Sheila Reinehr

Graduate Program in Computer Science, Pontifical Catholic University of Paraná State, No. 1155, Imaculada Conceição Avenue, Curitiba, Brazil

ARTICLE INFO

Keywords:

Ontology
Organizational learning
Learning organization

ABSTRACT

Although ontologies and organizational learning are issues that have been discussed for many years, there is not an approach on literature that gives an overview about how both issues have been applied together. This literature review has the objective of exploring how ontologies are being applied in the organizational learning process recently; as a consequence, only studies from the year of 2005 onwards have been searched. The identification process produced 353 papers from 11 different databases. After applying the exclusion criteria, the set was reduced to 11 papers, which clearly fitted to the criteria defined for accomplishment of the systematic review, which were then analyzed and classified. The papers have been classified according to the structure and level of the ontologies. Furthermore, the Information Technology (IT) used in conjunction with ontology was identified, as well as the way ontologies and IT can act as a means of facilitating the organizational learning process. It was observed that although ontologies are rather important, a very few number of researches have applied ontologies in the organizational learning processes. In a general way, ontologies and IT encourage the sharing of knowledge and formalization.

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Exemplo de RSL

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Exemplo de RSL

Retrospective and Trends in Requirement Engineering through WER

Joselaine Valaski, Wilian Stancke, Sheila Reinehr, Andreia Malucelli

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joselaine.valaski@pucpr.br, stancke@ieee.org,
sheila.reinehr@pucpr.br, malu@pucpr.br

Abstract. This work refers to the review of 258 papers published in the WER throughout 15 editions. This review's goal was to identify the most active research groups within this workshop, the most debated topics and the trends in the Requirements Engineering area. The results showed that Brazil, Argentina and Spain hold the most active groups. Moreover, the results pointed out the requirements modeling as one of the most discussed topics in this event.

Keywords. Retrospective, Trends, Requirements Engineering, WER

Exemplo de RSL

Abstract. This work refers to the review of 258 papers published in the WER throughout 15 editions. This review's goal was to identify the most active re-

Valaski, J., Stancke, W., Reinehr, S., Malucelli, A. **Retrospective and Trends in Requirements Engineering through the WER.** Workshop em Engenharia de Requisitos (WER 2013), Montevideo. Memorias..., 2013. p.135 - 148

MAPEAMENTO

MAPEAMENTO SISTEMÁTICO
É MUITO USADO PARA
MAPEAR UMA ÁREA DO
CONHECIMENTO MAS DE
UMA FORMA MAIS
GENÉRICA.



RSL

REVISÃO SISTEMÁTICA
DA LITERATURA É MAIS
USADA DEVIDO AO SEU
RIGOR CIENTÍFICO E POR
APROFUNDAR AS
QUESTÕES.

Meta Análise

- “Uma forma de estudo secundário onde a síntese da pesquisa é baseada em métodos quantitativos estatísticos”

Kitchenham, B. Guidelines for Performing Systematic Reviews in Software Engineering. EBSE Technical Report. 2007

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Fases para RSL



Planejar a Revisão



Conduzir a Revisão



Documentar a Revisão

- Identificar claramente a necessidade da RSL
- Procurar por outras RSL sobre o tema
 - Avaliar a qualidade das RSL existentes
- Desenvolver o protocolo de revisão

- O protocolo para a revisão sistemática deve especificar o processo e os métodos que serão aplicados.
- O protocolo inclui:
 - as questões de pesquisa
 - a estratégia de busca
 - os critérios de inclusão e exclusão

- Sobre a construção do protocolo:
 - Um mapeamento sistemático prévio pode ajudar a definir as questões de pesquisa da RSL
 - Questões novas podem surgir na medida que a pesquisa avança
 - Todos os membros da pesquisa devem participar da construção do protocolo para compreender os procedimentos de extração
 - Realizar projeto piloto para testar o protocolo

Planning questions	Planning answers
What are the review's objectives?	Identify the existent sustainable software engineering practices applied in the industry and academy to support the theoretical framework of case study.
What sources were searched to identify primary studies? Were there any restrictions?	Scopus, IEEEExplorer, Science Direct and ACM.
What were the inclusion/exclusion criteria and how were they applied?	Inclusion Criteria: I1 - Papers from Green IN Software, Sustainable Software Engineering, Green Software Engineering Conferences and Journals. I2 - Papers that practices of sustainable software engineering are clearly defined in the results and discussion of primary studies I3 - Papers related do sustainable software engineering with proof of concept in the context of industry and academy application. Exclusion Criteria: E1 - Discard of duplicated papers. E2 – Results that are only conference, book chapter description, call for papers, special issue, editorial E3 – Papers from conferences and journals that are related to other areas of knowledge. E4 – Papers from conferences and journals that are not related to Software Engineering. E5 – Papers related to Software Engineering by not related to Green IN Software Engineering. E6 - Papers that has not proven the empirical purpose in the context of industry and academy application.
What criteria were used to assess the quality of primary studies?	All primary studies must be related to green software engineering, sustainable software engineering, greener software and green in software engineering. In addition, it needs to report a case study, industry validity, academic validity, or proposed model evaluation in the industry in any stages of software development life cycle.
How were quality criteria applied?	• Assess the risk of bias in included studies caused by inadequacies in study design, conduct or analysis that may have led to the effect being over or underestimated. • The level of detail required in the assessment. • The ability to distinguish between internal validity (risk of bias) and external validity (generalizability) • Consider individual aspects of methodological quality in the quality assessment and synthesis. • Consider the potential impact that methodological quality had on the findings of the included.
How were the data extracted from the primary studies?	Papers were extracted by executing the string search in the scientific databases.
How were the data synthesized?	The data is synthesized is based on Inclusion and Exclusion Criteria.
How were differences between studies investigated?	Evaluation presented about proposed theory in the industry.
How were the data combined?	Grounded theory is used to combine the studies, create categories and define the practices to be identified in the cases study.

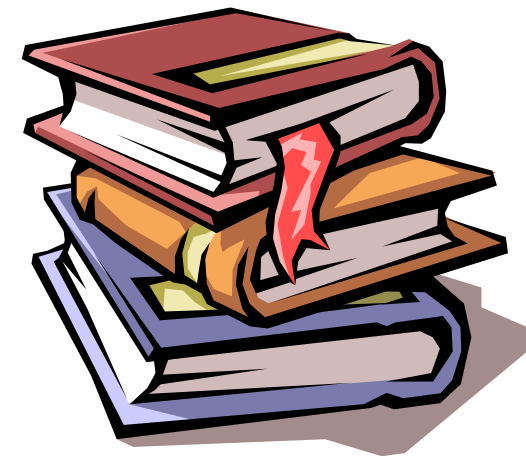
Planejar a Revisão
Exemplo

- Identificar a pesquisa
- Selecionar os estudos primários
- Analisar a qualidade dos artigos encontrados
- Classificar os artigos
- Sintetizar os dados

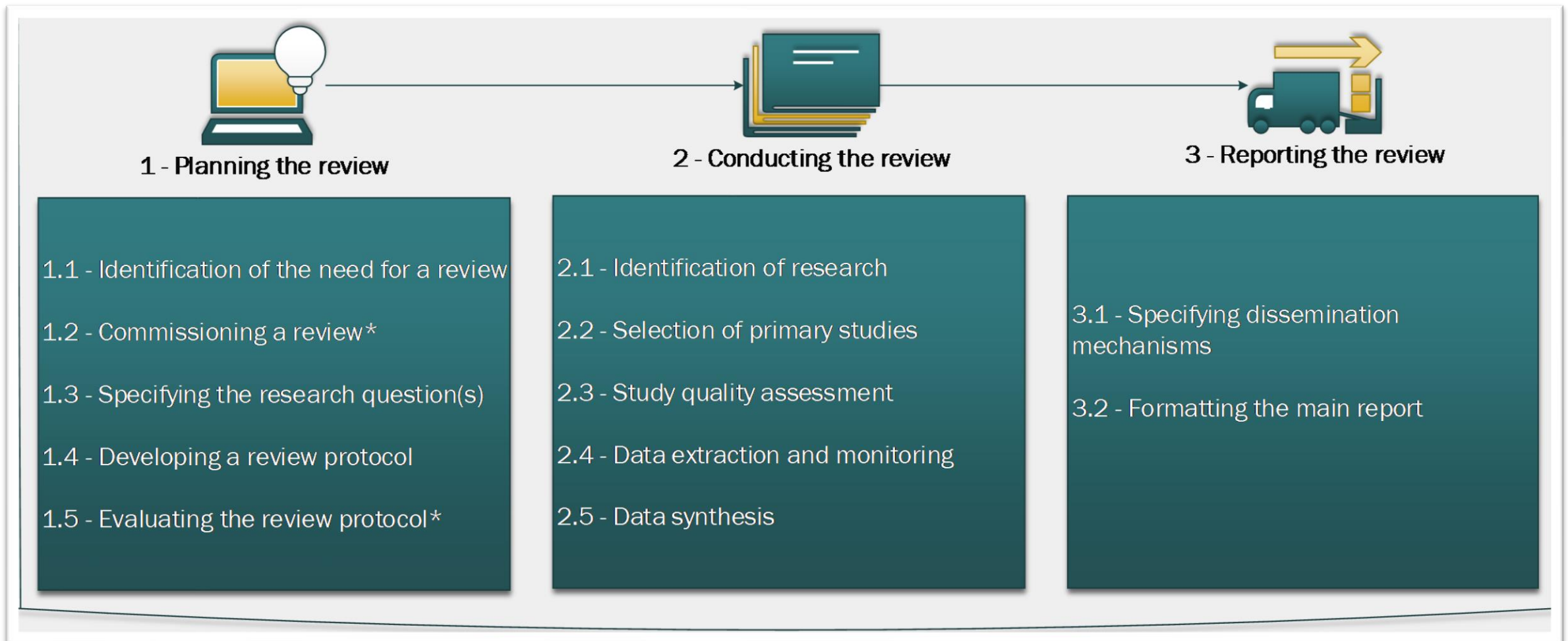
Fases da RSL

Documentar a Revisão

- Organizar o documento de RSL de acordo com o veículo de publicação: artigo, dissertação ou tese.



Exemplo



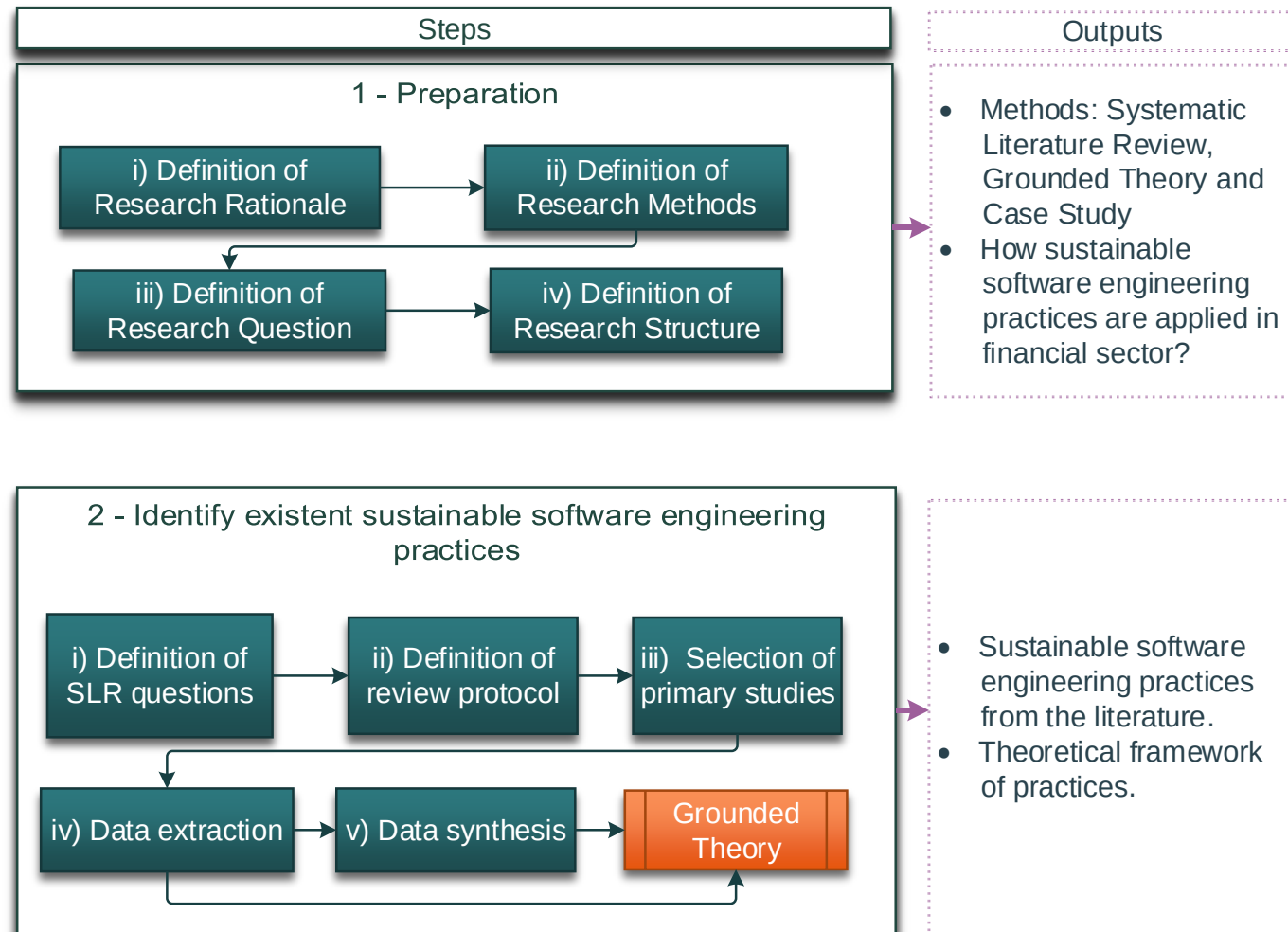
Agenda

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Passos para a RSL

- 1 • Definir objetivo/questões da revisão
- 2 • Definir palavras chaves
- 3 • Selecionar as bases para pesquisa
- 4 • Definir período e tipo de publicação
- 5 • Definir campos de busca
- 6 • Definir os critérios de inclusão e exclusão
- 7 • Definir os critérios de qualidade
- 8 • Executar as pesquisas e filtrar artigos
- 9 • Tabular resultados
- 10 • Ler, fazer resumos e anotações
- 11 • Classificar os artigos Seleccionados
- 12 • Organizar o documento

Exemplo de condução



1. Definir objetivo e questões

- Fase mais importante.
- Verificar...

A pesquisa é importante para os profissionais e pesquisadores ?

Resultará em mudanças na prática/pesquisas, ou aumentará a confiança nas práticas/pesquisas atuais ?

Identifica discrepâncias entre crenças e realidades ?

1. Definir objetivo e questões

Por que alguém
teria interesse em
ler a sua RSL?



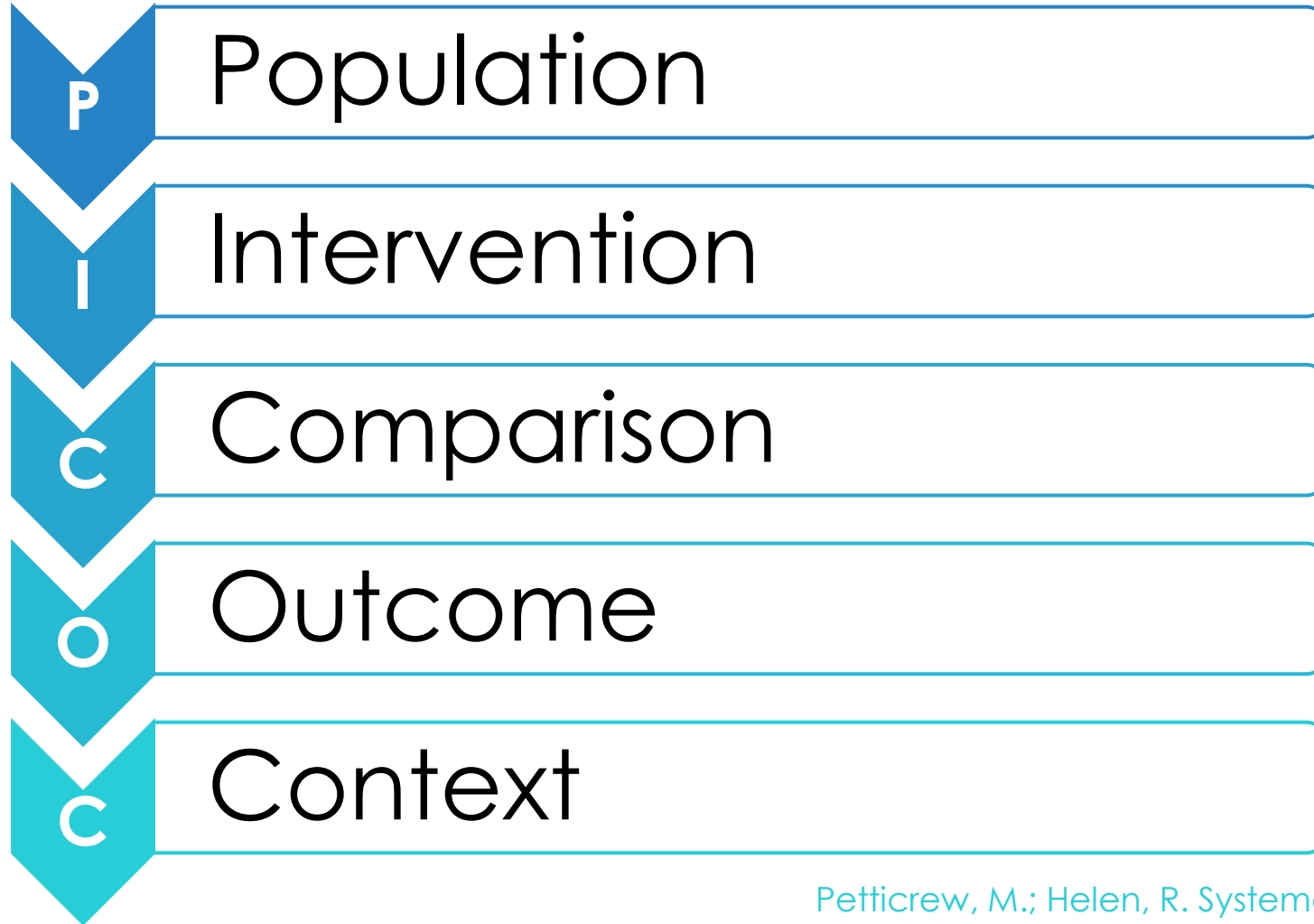
1. Definir objetivo (Exemplo)

The goal of this paper is to identify and understand the main contributions of the state-of-art towards software engineering in startups. To this end, we perform a systematic mapping study (SMS) [5,15] aimed at:

- characterizing the state-of-art research on startups,
- understanding the context that characterizes startups,
- determining the potential for technology transfer of the state-of-art research on startups,
- extracting and analyzing software development work practices used in startups.

Paternoster, N.; Giardino, C.; Unterkalmsteiner, M.; Gorschek, T.; Abrahamsson, P. Software development in startup companies: A systematic mapping study. Information

1. Definir questões de pesquisa



1. Definir questões de pesquisa

■ Population

- A população pode ser:
 - Um papel específico (gerente, analista, testador etc.)
 - Uma categoria (experiente, junior etc.)
 - Uma área de aplicação
 - Um grupo industrial (telecomunicações, pequenas empresas de TI etc.)

Petticrew, M.; Helen, R. Systematic Reviews in the Social Sciences: A Practical Guide, Blackwell Publishing

1. Definir questões de pesquisa

■ Intervention

- A intervenção pode ser:
 - Tecnologia
 - Método
 - Ferramenta
 - Etc.
- Exemplo: Ferramentas para Engenharia de Requisitos, Método de Classificação

Petticrew, M.; Helen, R. Systematic Reviews in the Social Sciences: A Practical Guide, Blackwell Publishing

1. Definir questões de pesquisa

■ Comparison

- Em Engenharia de Software pode ser método/tecnologia/ferramenta/procedimento com o qual se deseja realizar a comparação
- Exemplo: comparar pessoas que tiveram treinamento com pessoas que não tiveram treinamento em uma ferramenta.

Petticrew, M.; Helen, R. Systematic Reviews in the Social Sciences: A Practical Guide, Blackwell

1. Definir questões de pesquisa

■ Outcome

- Em Engenharia de Software pode ser o fator de importância para o praticante
- Exemplo: maior produtividade, custos reduzidos etc.

Petticrew, M.; Helen, R. Systematic Reviews in the Social Sciences: A Practical Guide, Blackwell Publishing

1. Definir questões de pesquisa

■ Context

- Refere-se ao contexto onde a comparação é realizada (academia, indústria, estudantes, profissionais etc.)
- É comum se realizar experimentos com estudantes, mas esta não é uma amostra representativa (pode-se escolher remover estes estudos da RSL)

Petticrew, M.; Helen, R. Systematic Reviews in the Social Sciences: A Practical Guide, Blackwell Publishing

1. Definir questões (Exemplo)

The research question driving this mapping study is: *What is the state-of-art in literature pertaining to software engineering in startups?* To answer this question, we state the following sub-questions:

- RQ1 What is the context that characterizes software development in startups?
- RQ2 To what extent does the research on startups provide reliable and transferable results to industry?
- RQ3 What are the reported work practices in association with software engineering in startups?

Paternoster, N.; Giardino, C.; Unterkalmsteiner, M.; Gorschek, T.; Abrahamsson, P. Software development in startup companies: A systematic mapping study. *Information and Software Technology*,

1. Definir questões (Exemplo)

Research Questions for the Systematic Review

ID	Question	Aim
RQ1	What types of evaluation strategies are used to evaluate SPI initiatives?	To identify which concrete evaluation strategies are used and how they are applied in practice to assess SPI initiatives.
RQ2	What are the reported metrics for evaluating the SPI initiatives?	To identify the metrics which are commonly collected and used to evaluate SPI initiatives.
RQ3	What measurement perspectives are used in the evaluation and to what extent are they associated with the identified SPI initiatives?	To determine from which measurement perspective SPI initiatives are evaluated. Furthermore, to analyze any relationship between SPI initiatives and measurement perspectives.
RQ4	What are the confounding factors in relation to the identified evaluation strategies?	To identify the reported factors that can distort and hence limit the validity of the results of the SPI evaluation. To determine if these issues are addressed and to identify possible remedies.

Unterkalmsteiner, M.; Gorschek, T.; Islam, A.K.M.; Cheng, C. K.; Permadi, R. B.; Feld, R. Evaluation and Measurement of Software Process Improvement - A Systematic Literature Review. IEEE Transactions on Software Engineering, v. 38, n.2 ,

1. Definir questões (Exemplo)

Tabela 1: Questões de pesquisa

Questão	Descrição
QP-1	Como requisitos são elicitados a partir de modelos de processos de negócios?
QP-1.1	Quais os tipos de requisitos que são elicitados?
QP-1.2	Em que forma os requisitos são elicitados?
QP-2	Quais técnicas de especificação de requisitos são utilizadas?

Unterkalmsteiner, M.; Gorschek, T.; Islam, A.K.M.; Cheng, C. K.; Permadi, R. B.; Feld, R. Evaluation and Measurement of Software Process Improvement - A Systematic Literature Review. IEEE Transactions on Software Engineering, v. 38, n.2 ,

1. Definir objetivo e questões

metrics research. It aims to identify and categorise influential software metrics research and also to consider the opportunity for aggregating research papers. Thus it addresses two research questions:

1. What papers are currently most influential in the software metrics research community?
2. Can software metrics papers on a similar topic be meaningfully aggregated?

2. Definir palavras chaves

- Selecionar todos os termos relevantes para a área pesquisada
- Buscar o maior número de artigos relevantes
- Verificar todos os sinônimos
- Fazer todas as combinações possíveis
- Pesquisar as palavras chaves utilizadas por conferências e journals da área
- Faça simulações/piloto em algumas bases científicas, conferências e/ou journals

2. Definir palavras chaves (Exemplos)

Tabela 2: *String* de busca padrão

((“business process model”) AND (“functional requirement” OR “non functional requirement” OR “Requirement Engineering” OR “Requirement Elicitation” OR “Elicitation of Requirement”) AND (BPMN OR “Business Process Model and Notation” OR “Business Process Modeling Notation” OR “Business Process Modelling Notation”))

2. Definir palavras chaves (Exemplos)

Core concepts	Terms
Software Startups	software startup*; software start-up*; early-stage firm; early-stage compan*; high-tech venture*; high-tech start-up*; start-up compan*; startup compan*; lean startup*; lean start-up*; software package start-up*; software package startup*; IT start-up*; IT startup*; software product startup*; software start up*; internet start-up*; internet startup*; web startup*; web start-up*; mobile startup*; mobile start-up*;
Development	develop*; engineer*; model*; construct*; implement*; cod*; creat*; build*;
Strategy	product*; service*; process*; methodolog*; tool*; method*; practice*; artifact*; artefact*; qualit*; ilit*; strateg*; software;

Paternoster, N.; Giardino, C.; Unterkalmsteiner, M.; Gorschek, T.; Abrahamsson, P. Software development in startup companies: A systematic mapping study. Information and Software Technology, v.56, n.10, pp. 1200-1218

2. Definir palavras chaves (Exemplos)

Search process.

Search Id	Goal	SEARCH string	Purpose
Search 1	Metrics papers 2005	(TITLE-ABS-KEY(software) AND TITLE-ABS-KEY(metrics OR measurement)) AND PUBYEAR IS 2005 AND SUBJAREA(COMP)	Comparison of least and most cited papers for classification and trend analysis Assessment of proposed classification scheme Trialling inclusion/exclusion criteria
Search 2A-2F	Metrics papers 2000-2005	TITLE-ABS-KEY(software) AND TITLE-ABS-KEY(metrics OR measurement)) AND PUBYEAR IS X AND SUBJAREA(COMP) where X identifies the specific year	Identification of most cited papers for classification and trend analysis

Kitchenham, B. What's up with software metrics? A preliminary mapping study. The Journal of Systems and Software, v.83, 2010, pp.27-51

2. Definir palavras chaves (Exemplos)

Search Keywords

Population	Intervention
"process improvement" OR "process enhancement" OR "process innovation" OR SPI	measur* OR metric* OR success* OR evaluat* OR assess* OR roi OR investment* OR value* OR cost* OR effect* OR goal* OR result*

Unterkalmsteiner, M.; Gorschek, T.; Islam, A.K.M.; Cheng, C. K.; Permadi, R. B.; Feld, R.
Evaluation and Measurement of Software Process Improvement - A Systematic

3. Selecionar as bases para a pesquisa

- Selecionar bases de acordo com a área de pesquisa:
 - ScienceDirect
 - Scopus
 - SpringerLink
 - ACM Digital
 - IEEE
- As buscas direto nas bases são mais trabalhosas, mas fornecem resultados mais confiáveis que o uso do portal de periódicos da CAPES.
- Pode ser necessário alterar a string de busca de acordo com a base pesquisada.

3. Selecionar as bases para a pesquisa

Interesse: Uso das Ontologias na Aprendizagem Organizacional

Tipo de Conteúdo: Journal

Período: 2005-2010

■ Exemplo

Palavras em Qualquer Parte do Texto			Springer Link	ACM Digital Library	Science Direct	IEEE xplore
Learning Organization						
	Ontology		18	0	46	2
Organizational Learning						
	Ontology		37	0	132	9

Palavras em Qualquer Parte do Texto			General Science Full-Text (Wilson)	ArXiv.org	Cambridge	IOPscience (Institute of Physics - IOP)	Oxford Journals (Oxford University Press)	Science (AAAS)	Wiley Online Library
Learning Organization									
	Ontology		0	0	4	0	1	0	67
Organizational Learning									
	Ontology		0	5	4	0	2	0	111

Bases principais para assuntos interdisciplinares

3. Selecionar as bases para a pesquisa

Selected databases and retrieved papers.

ID	Database	Papers
A	Inspec/Compendex (www.engineeringvillage2.org)	640
B	IEEE Xplore (ieeexplore.ieee.org)	132
C	Scopus (www.scopus.com)	468
D	ISI Web of Science (wokinfo.com)	293
E	ACM Digital Library (dl.acm.org/advsearch.cfm)	78
F	Google Scholar (scholar.google.com)	158
	Total	1769

Paternoster, N.; Giardino, C.; Unterkalmsteiner, M.; Gorschek, T.; Abrahamsson, P. Software development in startup companies: A systematic mapping study. Information and

3. Selecionar as bases para a pesquisa

Unterkalmsteiner, M.; Gorschek, T.; Islam, A.K.M.; Cheng, C. K.; Permadi, R. B.; Feld, R.
Evaluation and Measurement of Software Process

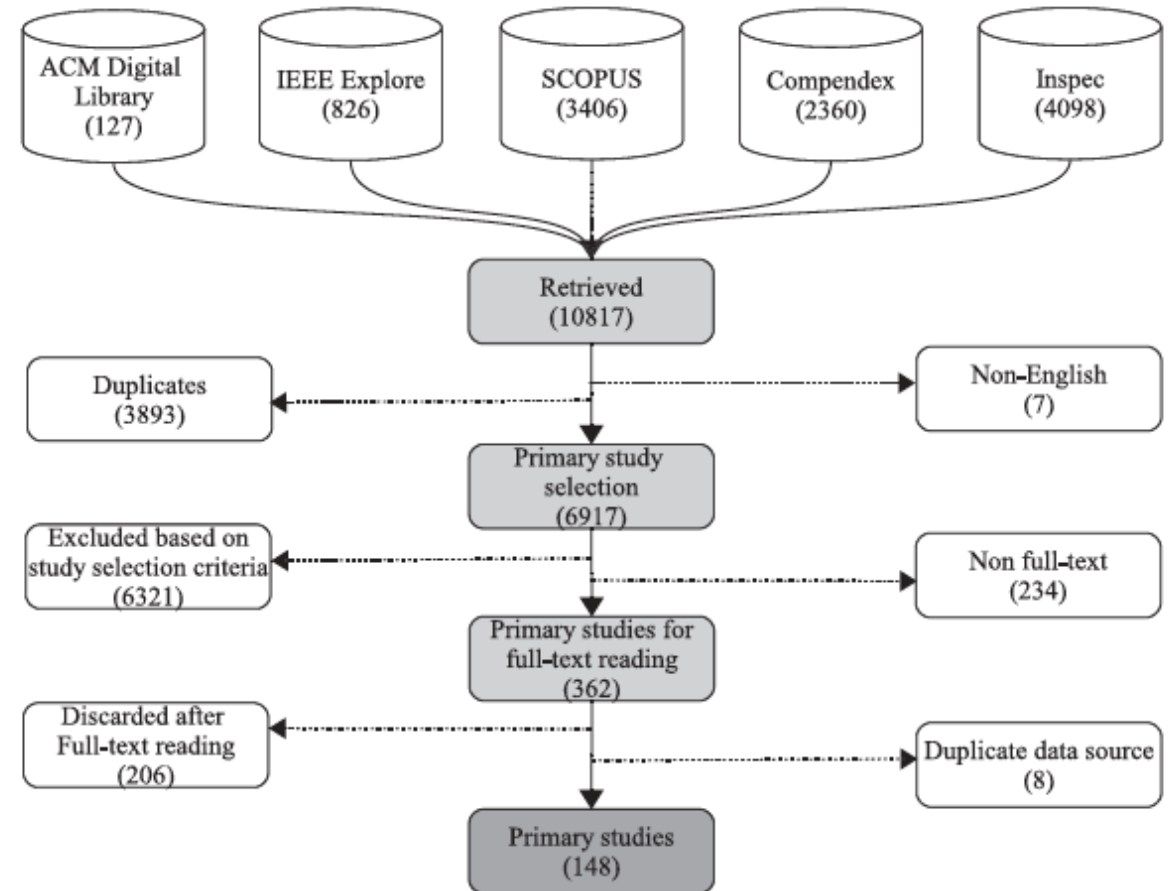


Fig. 3. Primary studies selection.

4. Definir período e tipo de publicação

- Período:
 - definir de acordo com a área
 - discutir com o orientador o período relevante
 - justificar a não inclusão de alguns períodos.
 - Por exemplo: últimos 10 anos (de 2005 – 2015)
- Tipo de publicação:
 - *journal* é o principal.
 - Verificar se é relevante incluir: *book, proceeding, newsletter* etc.

4. Definir período e tipo de publicação

I used SCOPUS to search for software metrics papers published in 2005 and the most cited software metrics papers in the years 2000–2005. The search processes started on 16th September 2008 and ended on 26th September 2008. The search process is outlined in Table 1.

Kitchenham, B. What's up with software metrics? A preliminary mapping study. *The Journal of Systems and Software*, v.83, 2010, pp.27-51.

5. Definir campos de busca

- Fulltext:
 - é o mais confiável, porém retorna grande quantidade de artigos, dependendo do escopo da pesquisa
- Abstract:
 - pode ser usado em escopo bem definido

QUIZ
MENTI.COM



6 – Definir os critérios de inclusão e exclusão

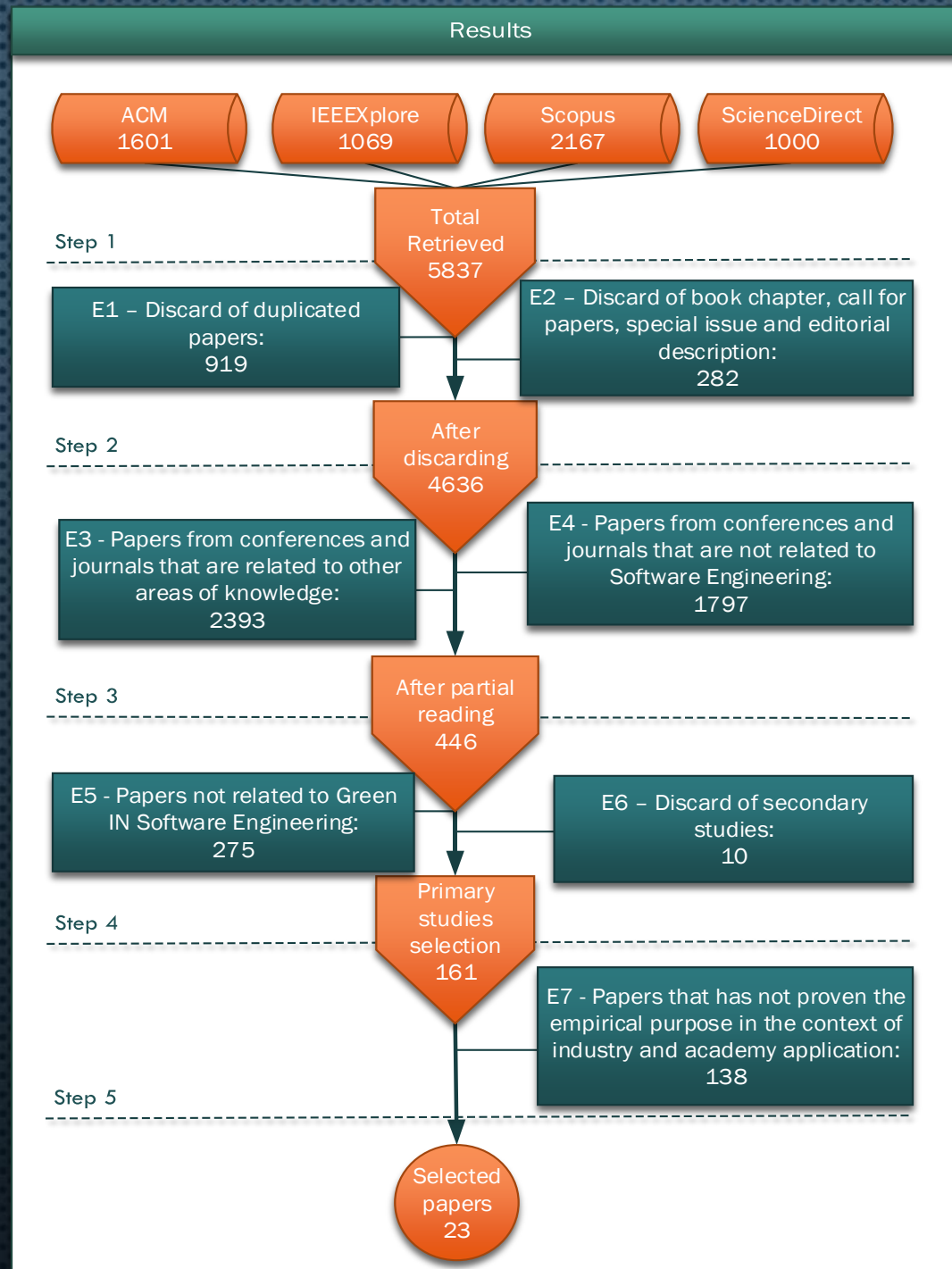
- Os critérios devem ser definidos de acordo com o objetivo do artigo.
- Exemplos de critérios de exclusão:
 - Artigos duplicados: encontrados em mais de uma base
 - *Grey Literature*: livros, teses, dissertações, etc.
 - Artigos redundantes: artigos diferentes mas com resultados similares
 - Foi encontrada a palavra chave mas não está no contexto do objetivo do artigo
 - ...

Tabela 3: Critérios de inclusão e exclusão

Sigla	Critério
CI1	O estudo apresenta um abordagem para elicitar requisitos a partir de modelos de processos de negócio em BPMN;
CI2	O estudo descreve uma ou mais atividades para elicitar requisitos de modelos de processos de negócio em BPMN;
CI3	O estudo apresenta uma adaptação do modelo de processos de negócio em BPMN para elicitar requisitos;
CE1	O estudo está fora do escopo de elicitação de requisitos a partir de modelos de processos de negócio em BPMN;
CE2	O estudo é um texto não revisado por pares; e
CE3	O estudo não está escrito em inglês ou português.

Acervo digital	Inicial	Duplicados	CE1	CE2	CE3	Seleção 1	Leitura	Seleção 2
ACM	122	-2	-110	-2	0	8	-8	0
Scopus	35	-7	-16	-4	-1	7	-4	3
Compendex	16	0	-9	0	0	7	-6	1
Science Direct	11	-1	-6	0	0	4	-4	0
IEEE Xplore	191	-4	-165	-3	0	19	-17	2
BTD	69	0	-60	0	0	9	-5	4
Total	444	-14	-366	-9	-1	54	-44	10

6 – DEFINIR OS CRITÉRIOS DE INCLUSÃO E EXCLUSÃO EXEMPLO



6 – DEFINIR OS CRITÉRIOS DE INCLUSÃO E EXCLUSÃO EXEMPLO

7 – Definir os critérios de qualidade

- Avaliar a qualidade e credibilidade dos estudos primários para dar suporte ao processo de inclusão/exclusão e atribuir pesos para os estudos durante a síntese.
- Não há um método padrão para atribuir qualidade mas o objetivo é diminuir o viés e maximizar a validação.

7 – Definir os critérios de qualidade

- Exemplos de critérios de qualidade:
 - O estudo foi avaliado empiricamente ?
 - As limitações do estudo foram discutidas ?
 - Houve aplicação na indústria ?
 - Objetivo claramente definido ?
- Algumas revisões têm reutilizado critérios definidos por outros autores: Guyatt et al. (2008); Dybä, T. & Dingsoyr (2008); Achimugu et al. (2014); Kitchenham & Brereton (2013), etc.

8. Executar as pesquisas e filtrar artigos

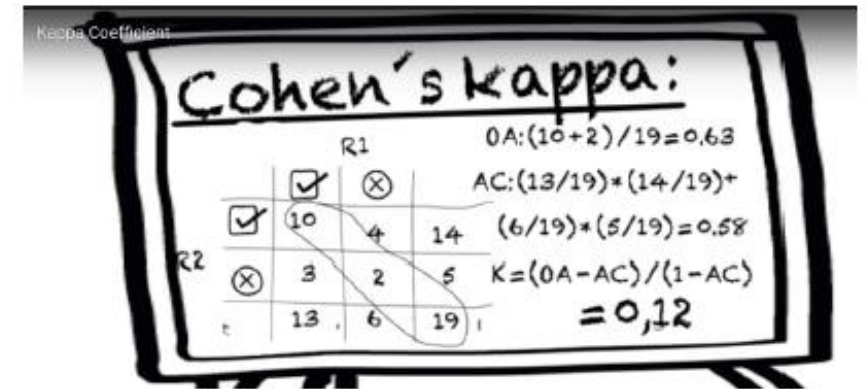
- **Registrar** a quantidade de artigos encontrados.
 - Algumas bases (ScienceDirect) permitem gravar as consultas realizadas
- **Filtrar** artigos no **mínimo** pela leitura do *abstract*.
 - Filtrar apenas pelo título é “perigoso”
- Quando o *abstract* gerar **dúvidas**, procurar ler pontos específicos do artigo.
 - Ex: buscar por uma das palavras chaves no texto e verificar o que diz a respeito do termo. Pode ser que ela seja encontrada apenas na referência.
- Quando houver dúvidas, separar artigo para leitura completa.

8. Executar as pesquisas e filtrar artigos

- Sempre que possível minimamente 2 pesquisadores devem participar da seleção e leitura dos artigos, e se necessário um terceiro pesquisador para situação de empate.
- Cuidar para que o entendimento e envolvimento de todos seja igual. Critérios de inclusão e exclusão bem definidos fazem a diferença.

8. Executar as pesquisas e filtrar artigos

- Usar o coeficiente de Kappa, que mede o grau de concordância das avaliações feitas por diversos avaliadores.



Kappa Statistic

<0.00
0.00-0.20
0.21-0.40
0.41-0.60
0.61-0.80
0.81-1.00

Strength of Agreement

Poor
Slight
Fair
Moderate
Substantial
Almost Perfect

9. Tabela os resultados

- Não há padrão.
- Não há ferramenta específica para isso.
- Seja o mais organizado possível !

Lembrar que serão
centenas de artigos !!

centenas de artigos !!
lembrar que serão



9. Tabela os resultados

- Exemplo com uso do excel:

	C11		f _x	Ontology	
	A	B	C		
1	Fonte	Palavra 1	Palavra 2	Artigo	
2	ScienceDirect	Organizational Learning	Ontology	A business process context for Knowledge Management	
3	IEEE	Organizational Learning	Ontology	A Cognitive Model of Improvisation in Emergency Management	
4	Wiley	Organizational Learning	Ontology	A comparative study on e-learning technologies and products: from	
5	Wiley	Learning Organization	Ontology	A comparative study on e-learning technologies and products: from the East to the V	
6	ScienceDirect	Organizational Learning	Ontology	A conceptual framework for analysing adaptive capacity and multi-level learning proce	
7	ScienceDirect	Learning Organization	Ontology	A conceptual model of personalized virtual learning environments	
8	ScienceDirect	Organizational Learning	Ontology	A context-based model for Knowledge Management embodied in work processes	
9	ScienceDirect	Organizational Learning	Ontology	A contingent approach on knowledge portal design for R&D teams: Relative importan	
10	ScienceDirect	Organizational Learning	Ontology	A cyber-crime investigation framework	
11	ScienceDirect	Organizational Learning	Ontology	A Delphi study of knowledge management systems: Scope and requirements	
12	Wiley	Learning Organization	Ontology	A DOOYEWEERD-BASED APPROACH TO REGIONAL ECONOMIC	
13	SpringerLink	Organizational Learning	Ontology	A framework for designing the enterprise-wide e-commerce portal for evolving organiz	
14	ScienceDirect	Learning Organization	Ontology	A framework for examining leadership in extreme contexts	
15	ScienceDirect	Organizational Learning	Ontology	A framework for examining leadership in extreme contexts	
16	IEEE	Organizational Learning	Ontology	A generic library of problem solving methods for scheduling applications	
17	IEEE	Organizational Learning	Ontology	A knowledge creation info-structure to acquire and crystallize the tacit knowledge of l	
18	Wiley	Organizational Learning	Ontology	A knowledge management model: Implications for enhancing quality	
19	Wiley	Learning Organization	Ontology	A knowledge management model: Implications for enhancing quality in health care	
20	Wiley	Organizational Learning	Ontology	A knowledge management technology architecture for educational	
21	SpringerLink	Learning Organization	Ontology	A Knowledge-Based Approach to Environmental Biomonitoring	
22	ScienceDirect	Organizational Learning	Ontology	A Levinasian ethics critique of the role of management and control systems by large	

10. Ler, fazer resumos e anotações

- Não há padrão
- Sugestão: usar Mendeley

The screenshot displays the Mendeley Desktop application. On the left, the 'My Library' pane shows a list of folders, with '_Artigo_EstiloAprendizagemMaterial' selected. Below this, a 'Filter by Publications' dropdown is set to 'All'. The main pane on the right shows a list of articles under the title '_Artigo_EstiloAprendizagemMaterial'. The table below represents the data shown in this list.

★	●	📄	Authors	Title	Year	Published In	Added
☆	•	📄	Beacham, N.A.; ...	An investigation into the effects that digital media can have on the learning outcomes of individuals who have dyslexia	2006	Computers & Education	jan 17
☆	•	📄	Tseng, J.C.R.; C...	Development of an adaptive learning system with two sources of personalization information	2008	Computers & Education	jan 17
☆	•	📄	Ozpolat, E.; Akar...	Automatic detection of learning styles for an e-learning system	2009	Computers & Education	jan 17
☆	•	📄	Holzinger, A.; Kic...	Learning performance with interactive simulations in medical education: Lessons learned from results of learning compl...	2009	Computers & Education	jan 17
☆	•	📄	Graf, Sabine; Liu...	Learning styles and cognitive traits-Their relationship and its benefits in web-based educational systems	2009	Computers in Human Behavior	jan 17
☆	•	📄	Essalmi, Fathi; Ay...	A fully personalization strategy of E-learning scenarios	2010	Computers in Human Behavior	jan 17
☆	•	📄	Liegle, J; Janicki, T	The effect of learning styles on the navigation needs of Web-based learners	2006	Computers in Human Behavior	jan 17
☆	•	📄	Pan, Weidong; H...	Facilitating knowledge construction by customized supportive services	2007	Education and Information Tech...	jan 17
☆	•	📄	Liu, Tzu-Chien; G...	Coping with mismatched courses: students' behaviour and performance in courses mismatched to their learning styles	2009	Educational Technology Rese...	jan 17
☆	•	📄	Gamalel-Din, S.A.	Smart e-Learning: A greater perspective; from the fourth to the fifth generation e-learning	2010	Egyptian Informatics Jour...	26/10/1
☆	•	📄	Wang, T; Wang,...	Using a style-based ant colony system for adaptive learning	2008	Expert Systems with Applications	jan 17
☆	•	📄	Yang, Y.J.; Wu, C	An attribute-based ant colony system for adaptive learning object recommendation	2009	Expert Systems with Applications	26/10/1
☆	•	📄	Yaghmaie, Mahka...	A context-aware adaptive learning system using agents	2010	Expert Systems with Applications	26/10/1
☆	•	📄	Wang, H; Li, T; C...	A web-based tutoring system with styles-matching strategy for spatial geometric transformation	2006	Interacting with Computers	jan 17
☆	•	📄	Santos, O.C.; Bo...	Modeling recommendations for the educational domain	2010	Procedia Computer Science	jan 17

10. Ler, fazer resumos e anotações

★	●	📄	Authors	Title	Year	Publishe	Ac
☆	●	📄	Beac...	An investigation into the effects that digital media ...	2006	Computers & Educati...	ja
☆	●	📄	Tseng...	Development of an adaptive learning system ...	2008	Computers & Educati...	ja
☆	●	📄	Ozpol...	Automatic detection of learning styles for an e-l...	2009	Computers & Educati...	ja
☆	●	📄	Holzin...	Learning performance with interactive simulations in ...	2009	Computers & Educati...	ja
☆	●	📄	Graf, ...	Learning styles and cognitive traits-Their relat...	2009	Computers in Human ...	ja
☆	●	📄	Essal...	A fully personalization strategy of E-learning sce...	2010	Computers in Human ...	ja
☆	●	📄	Liegle...	The effect of learning styles on the navigation n...	2006	Computers in Human ...	ja
☆	●	📄	Pan, ...	Facilitating knowledge construction by customiz...	2007	Education and Infor...	ja
☆	●	📄	Liu, Tz...	Coping with mismatched courses: students' behavi...	2009	Educational Technolo...	ja
☆	●	📄	Gama...	Smart e-Learning: A greater perspective; fro...	2010	Egyptian Informati...	26
☆	●	📄	Wang...	Using a style-based ant colony system for adapti...	2008	Expert Systems ...	ja
☆	●	📄	Yang, ...	An attribute-based ant colony system for adapti...	2009	Expert Systems ...	26
★	●	📄	Yagh...	A context-aware adaptive learning system using age...	2010	Expert Systems ...	26
☆	●	📄	Wang...	A web-based tutoring	2006	Interacting	ja

Details

Notes

Notes:
B *I* U
Padrão: SCORM, IMS
Estilo de Aprendizagem: Kolb
Ontologia: Armazenamento, sequência e adaptação.
Armazenamento dos tópicos das aulas.
Usa mecanismos de inferência por meio da camada de negócios. Usa OWL.

Proposta de um sistema adaptativo utilizando agentes.
1 - Introdução
Sistemas adaptativos de aprendizagem tem sido utilizados para melhorar eficiência da aprendizagem disponibilizando os recursos de acordo com o estilo de aprendizagem. A idéia principal deste tipo de abordagem é que não existe um estilo de aprendizagem que se enquadre em todos os tipos de necessidades dos aprendizes.
Ao utilizar os sistemas adaptativos há duas abordagens, a primeira adequar os materiais com as necessidades individuais e a segunda disponibilizar materiais mais apropriados aos indivíduos de acordo com

11 – Classificar os Artigos Seleccionados

- Os resultados devem ser organizados de maneira a apresentar informação clara, útil e inovadora.
- As questões de pesquisa ajudam na organização dos resultados.
- Pensar na classificação antes da leitura ajuda na organização.
- Organização da classificação, posteriormente, em gráficos e tabelas é mais interessante do que resumo de cada trabalho.

12. Organizar o documento

- Seguir padrão exigido (artigo, dissertação ou tese).
- Fazer uma boa introdução e revisão de literatura (background).
- Descrever as etapas do protocolo com nível de detalhe suficiente.
- Verificar a melhor maneira de apresentar/explorar os resultados: tabelas, quadros, gráficos etc. Evitar que sejam apenas descritivos.
- Nas discussões e conclusões, rever as questões de pesquisa e mostrar que foram respondidas (e como).

Organização de Resultados (Exemplo)

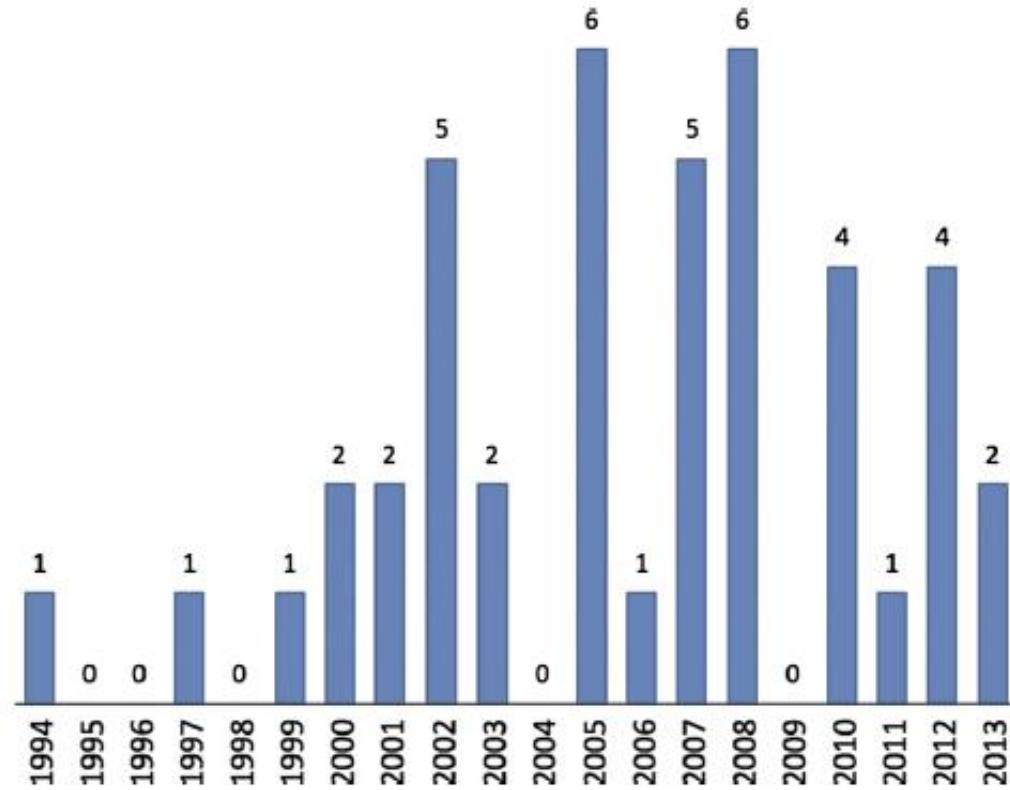
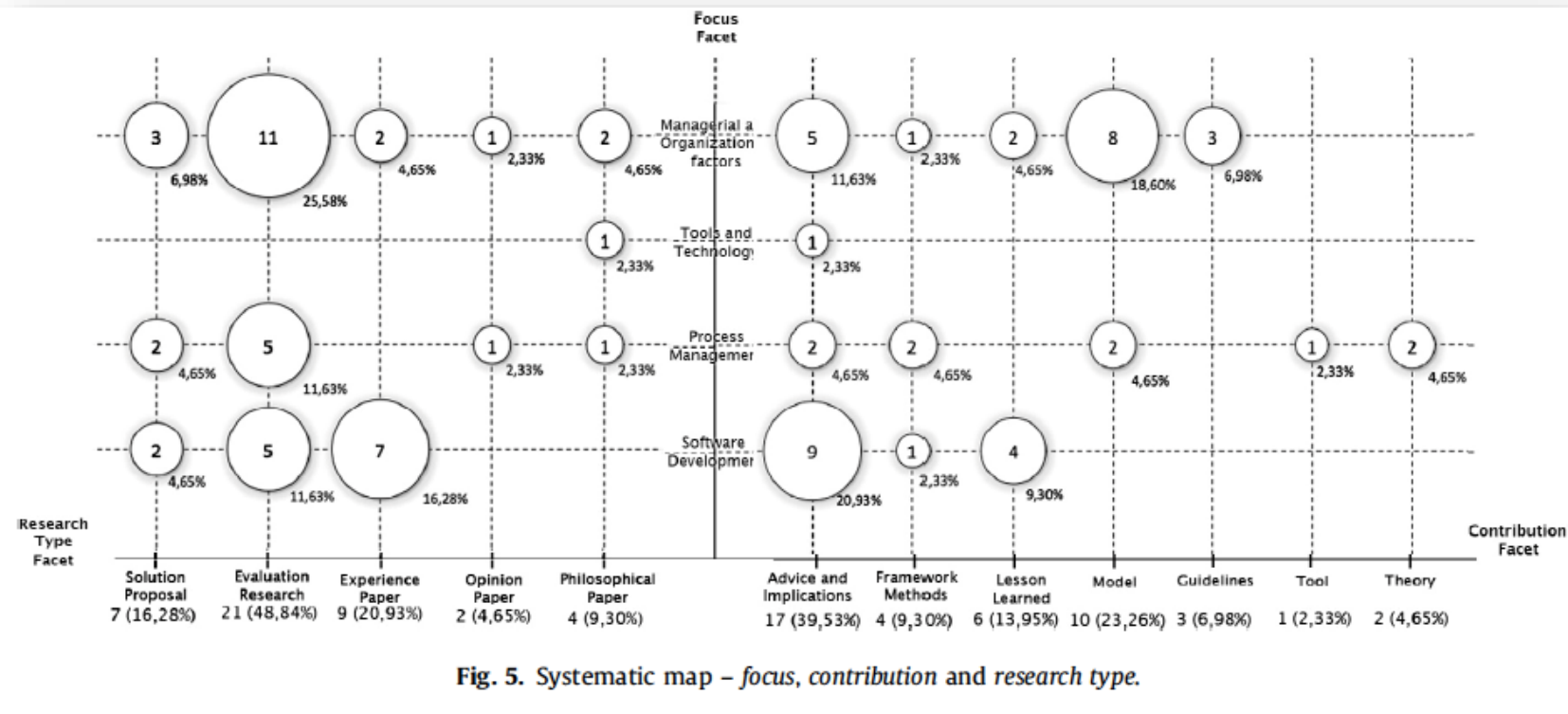


Fig. 4. Publication distribution-year.

Paternoster, N.; Giardino, C.; Unterkalmsteiner, M.; Gorschek, T.; Abrahamsson, P. Software development in startup companies: A systematic mapping study. Information and Software Technology

Organização de Resultados (Exemplo)



Paternoster, N.; Giardino, C.; Unterkalmsteiner, M.; Gorschek, T.; Abrahamsson, P. Software development in startup companies: A systematic mapping study. Information and Software

Organização de Resultados (Exemplo)

1st Author (year)	Research type	Contribution	Focus	Pertinence
Coleman (2008) [11]	Evaluation Research	Model	Process Management	Full
Kajko (2008) [13]	Evaluation Research	Model	Process Management	Full
Mater (2000) [68]	Evaluation Research	Model	Managerial & Organizational	Partial
Häsel (2010) [69]	Evaluation Research	Model	Managerial & Organizational	Marginal
Hanna (2010) [70]	Evaluation Research	Model	Managerial & Organizational	Marginal
Chorev (2006) [29]	Evaluation Research	Model	Managerial & Organizational	Marginal
Kakati (2003) [30]	Evaluation Research	Model	Managerial & Organizational	Marginal
Kim (2005) [71]	Evaluation Research	Model	Managerial & Organizational	Marginal
Coleman (2007) [72]	Evaluation Research	Theory	Process Management	Full
Coleman (2008) [24]	Evaluation Research	Theory	Process Management	Full
Bosch (2013) [73]	Evaluation Research	Framework & Methods	Process Management	Full
Midler (2008) [74]	Evaluation Research	Framework & Methods	Managerial & Organizational	Partial
Yogendra (2002) [75]	Evaluation Research	Guidelines	Managerial & Organizational	Partial
Yoffie (1999) [76]	Evaluation Research	Guidelines	Managerial & Organizational	Marginal
Camel (1994) [19]	Evaluation Research	Lesson Learned	Software Development	Full
Silva (2005) [67]	Evaluation Research	Lesson Learned	Software Development	Full
Jansen (2008) [77]	Evaluation Research	Lesson Learned	Software Development	Partial
Steenhuis (2008) [78]	Evaluation Research	Lesson Learned	Managerial & Organizational	Marginal
Lai (2010) [79]	Evaluation Research	Lesson Learned	Managerial & Organizational	Marginal
Tingling (2007) [80]	Evaluation Research	Advice & Implications	Software Development	Full
Li (2007) [81]	Evaluation Research	Advice & Implications	Software Development	Marginal
Blank (2013) [31]	Solution Proposal	Framework & Methods	Process Management	Partial
Zettel (2001) [82]	Solution Proposal	Framework & Methods	Software Development	Full
Crowne (2002) [8]	Solution Proposal	Advice & Implications	Software Development	Full
Mirel (2000) [83]	Solution Proposal	Advice & Implications	Managerial & Organizational	Partial

Paternoster, N.; Giardino, C.; Unterkalmsteiner, M.; Gorschek, T.; Abrahamsson, P. Software development in startup companies: A systematic mapping study. Information and

Organização de Resultados (Exemplo)

Journal	Number of papers
IEEE TSE	34
Journal of Systems and Software	9
Empirical Software Engineering	8
IEEE Software	5
Information and Software Technology	4
Computer	2
ACM Transactions on Software Engineering Methodology	2
IEE Proceedings Software	2
IEEE Trans on Parallel and Distributed Systems	2
IEEE Trans on Reliability	2
Others (1 paper only)	11
Number of different sources	20

Kitchenham, B. What's up with software metrics? A preliminary mapping study. The Journal of Systems and Software, v.83, 2010

Organização de Resultados (Exemplo)

Citations and paper type for the most cited papers.

Number of citations	Number of Journal papers	Number of Conference papers	Total
<10	10	2	12
10-19	31	11	41
20-29	17	4	21
30-39	8	2	10
40-49	10	0	10
50+	6	0	6
Total	81	19	100

Kitchenham, B. What's up with software metrics? A preliminary mapping study. The Journal of Systems and Software, v.83

Organização de Resultados (Exemplo)

Unterkalmsteiner, M.; Gorschek, T.;
Islam, A.K.M.; Cheng, C. K.;
Permadi, R. B.; Feld, R. Evaluation
and Measurement of Software
Process Improvement - A Systematic
Literature Review. IEEE Transactions
on Software Engineering, v. 38, n.2 ,
2012, pp.398-424 .

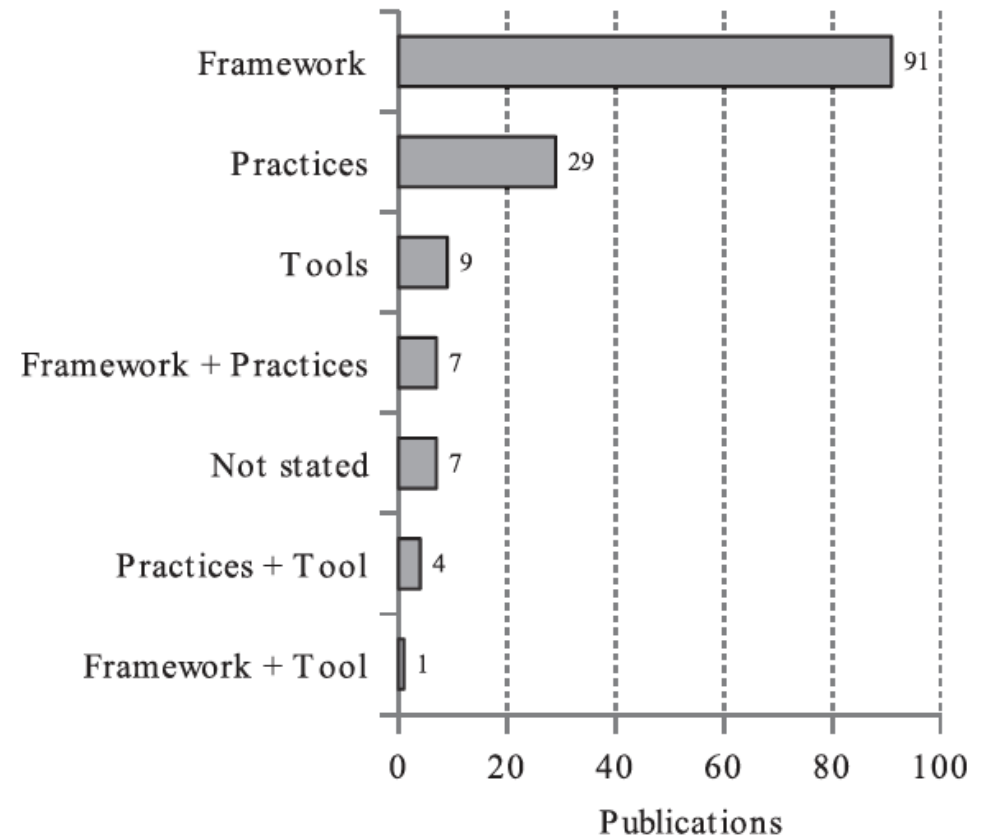


Fig. 5. SPI initiative distribution of the publications.

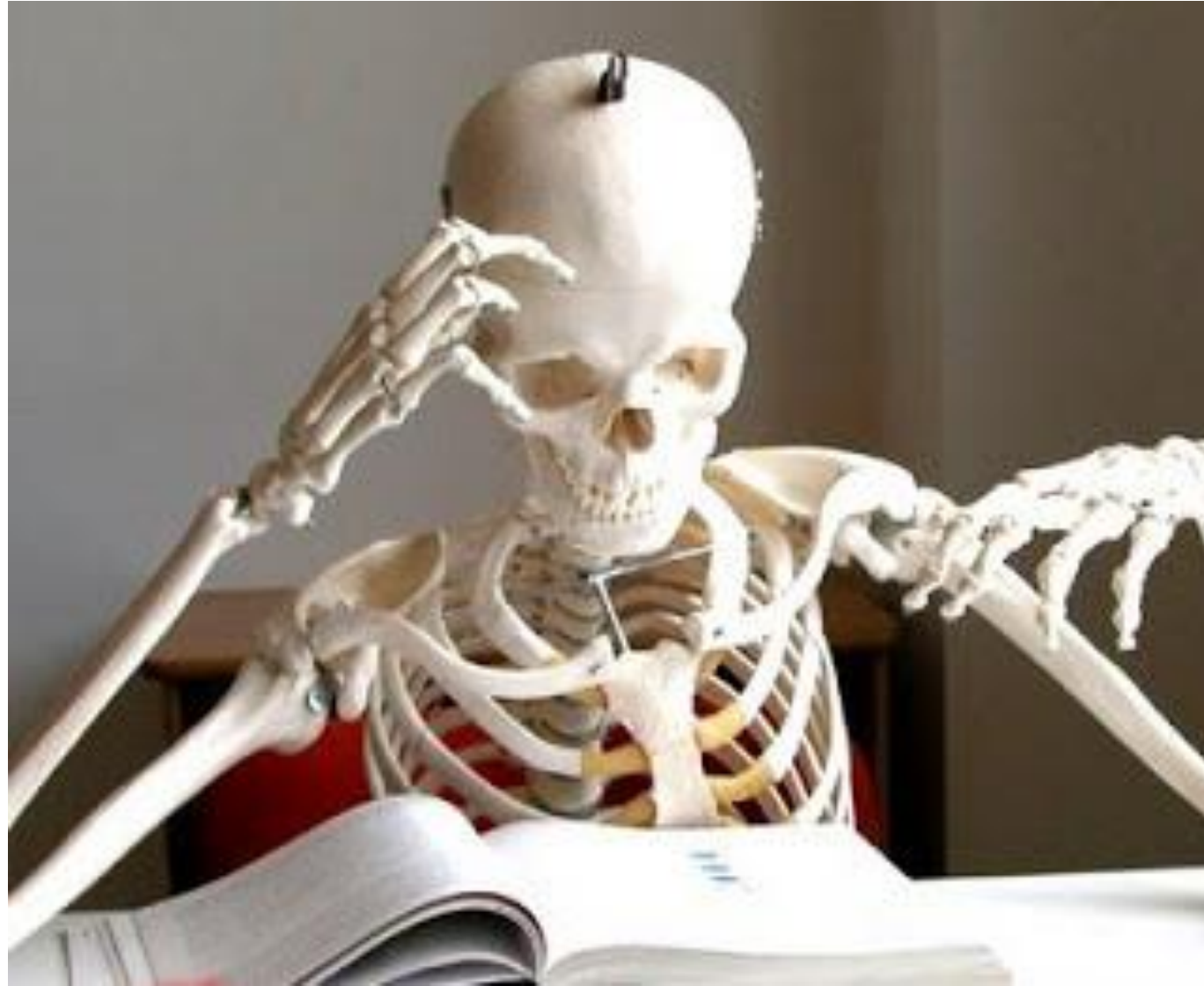
Organização de Resultados (Exemplo)

- Algumas questões que podem ser usadas para avaliar a qualidade dos estudos primários

Quality Assessment

ID	Quality assessment question	Yes	Partially	No
QA1	Is the aim of the research sufficiently explained?	138 (93.2%)	10 (6.8%)	0 (0.0%)
QA2	Is the presented idea/approach clearly explained?	115 (77.7%)	30 (20.3%)	3 (2.0%)
QA3	Are threats to validity taken into consideration?	16 (10.8%)	19 (12.8%)	113 (76.4%)
QA4	Is it clear in which context the research was carried out?	73 (49.3%)	54 (36.5%)	21 (14.2%)
QA5	Are the findings of the research clearly stated?	117 (79.0%)	30 (20.3%)	1 (0.7%)

RSL é para os FORTES !



Agenda

- Atividade 6 – Artigo RSL
 - Quiz Menti.com
- Visão geral
- Fases
- Condução
 - Quiz Menti.com
- Ferramentas
- Atividade 7 – Início mini pesquisa
- Próxima aula

ATIVIDADE 7:

Escolha um problema que pode ser investigado com uma RSL e defina:

- Objetivo da Pesquisa
- Questões de Pesquisa
- String Busca

CONDUÇÃO:

Individual/Duplas e Equipes

DURAÇÃO:

40min nos grupos do BB

2 minutos de apresentação

Individual/Duplas e Equipes

ENTREGA: Blackboard → Atividades Pedagógicas → Atividade Formativas → Atividade 7



Agenda

- Atividade 6 – Artigo RSL
 - Quiz Menti.com
- Visão geral
- Fases
- Condução
 - Quiz Menti.com
- Ferramentas
- Atividade 7 – Início mini pesquisa
- Próxima aula

Próxima aula...

- Condução dos passos da Revisão Sistemática da Literatura para começar a mini pesquisa.
- Acompanhamento individual, das duplas e equipes para esclarecimento de dúvidas.
- Aproveitar a aula para a mini pesquisa.

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